

Hands-On Magnetic Head Illusion: Sound Appears Inside the Head—But Only While Being Touched

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Abstract

We present the “Hands-On Magnetic Head Illusion,” a multisensory system in which the perceived location of sound is drawn toward the head—but only while the head is being touched. Building on prior studies, we hypothesize that this audiotactile magnet effect arises not merely from bodily configuration or the presence of touch, but from the perceptual and cognitive significance of the head as a body part. In a within-subjects experiment, we compared four tactile stimulation sites: (1) the top of the head, (2) a hand placed on the top of the head, (3) a hand placed on the shoulder, and (4) a hand positioned along the body’s midline in front of the head. Results showed that the illusion—i.e., the perceived shift of sound toward the point of contact—occurred strongly only when the contact involved the head or a hand on the head. In contrast, the effect was clearly weaker when touching the shoulder or a hand placed along the body’s midline, with the degree of perceived auditory compression being less than half of that in head-related conditions. These findings suggest that the illusion is specific to the head. Furthermore, the fact that direct skin contact was not necessary—as the illusion was also elicited when touching a hand placed on the head—indicates that the effect stems not from physical contact with the scalp, but rather from the perceptual activation of the head as a sensory stage.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

Keywords

Audio-tactile interaction; In-head localization;

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1 Introduction

In multisensory perception, tactile input has been shown to influence auditory spatial localization. For example, Caclin et al. (2002) [1] demonstrated that a tactile stimulus can “capture” the perceived location of a simultaneously presented sound, pulling it toward the touch point. Similarly, Tajadura-Jiménez et al. (2012) [3] showed that altering the spatial presentation of action-related sounds can recalibrate the perceived distance of touch on the body. These findings suggest that auditory localization is flexible and can be shaped by tactile and motor-related inputs.

In our previous work XRAYHEAD Garden [2], we demonstrated that when a sound is presented in sync with tactile stimulation to the head, the perceived location of the sound can be drawn inward, creating a vivid illusion of internal auditory localization. We referred to this phenomenon as the head-related audiotactile magnet effect. Interestingly, this effect occurred even in the absence of visual cues, suggesting that synchronous auditory-tactile input alone can cause external sounds to be perceptually “captured” inside the head.

Building on these findings, we hypothesize that the head has a unique tendency to attract perceived sound sources when touch is applied in synchrony with sound. Specifically, we propose that this phenomenon is not merely a function of spatial proximity, but rather reflects the perceptual and cognitive prominence of the head as the central stage of thought and awareness. This in-head



Figure 1: Typical Scenario showing the Hands-On Magnetic Head Effect: head touch draws moving sound perceptually toward the head under blocked vision.

attraction effect appears to be specific to the head, and does not generalize to other parts of the body—even those near the head or aligned along the body’s central axis.

In this demonstration, we present the Hands-On Magnetic Head Illusion, a system in which auditory localization is dynamically pulled into the head only while the head is being touched (Figure 1). This touch-gated magnetization effect illustrates how temporal tactile contact can momentarily transform the spatial experience of sound—without any visual involvement. We further report experimental results supporting the idea that this magnet effect is uniquely associated with the head region, rather than arising from spatial proximity or alignment along the body axis.

2 Experiment

2.1 Method

To examine whether the audiotactile magnet effect is specific to the head region, we conducted a within-subjects experiment comparing four tactile stimulation sites:

- (1) the top of the head,
- (2) the back of the hand placed on the head (Hand-on-Head),
- (3) the back of the hand placed on the shoulder (Hand-on-Shoulder)
- (4) the back of the hand placed in front of the head (Hand-in-Front-of-Head).

During each trial, participants wore a blindfold and received synchronized auditory and tactile stimulation for 30 seconds. The sound was generated using a light-reactive toy speaker, whose frequency was modulated in real time according to light intensity. The speaker was placed approximately 40 cm in front of the participant’s head. Tactile stimulation was delivered by pressing the designated body location with three fingers, modulated in sync with the audio.

After each trial, participants rated their experience on a 7-point scale in response to two questions: (1) whether the sound seemed to come from the touched location, and (2) whether they felt that their head or hand had moved forward. They also pointed to the perceived sound location on a horizontal reference board.

Twelve university students participated in the study.

2.2 Results

As shown in Figure 2, both subjective ratings and pointing responses indicated that the audiotactile magnet effect was significantly stronger when tactile stimulation was applied to the head or to a hand placed on the head (Hand-on-Head condition), compared to the other conditions. Participants more frequently reported that the sound seemed to originate from the point of contact, and pointed significantly closer to their head in these two conditions.

In contrast, the illusion was consistently weaker when the hand was placed on the shoulder or held in front of the head. Notably, even in the Hands-in-Front-of-Head condition—where the hand was spatially close but not touching the head—the perceived sound remained farther away. This suggests that mere proximity is not sufficient to induce the illusion.

Statistical analyses confirmed a significant difference between head-related conditions (Head and Hand-on-Head) and non-head conditions (Shoulder and Hands-in-Front-of-Head), indicating that

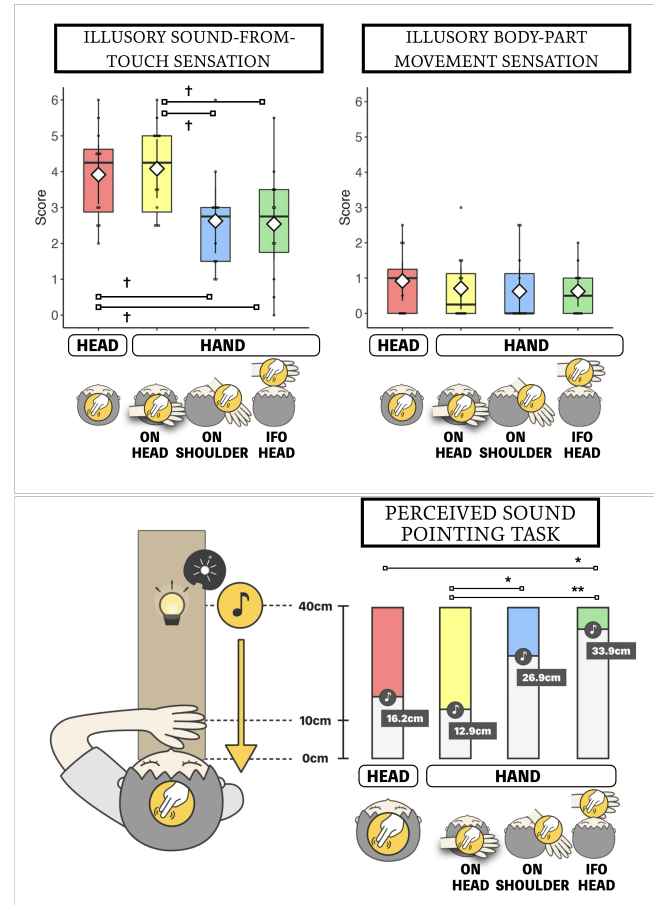


Figure 2: Experimental Results

the effect is specific to the head region. These findings support the hypothesis that the head’s perceptual and cognitive salience plays a key role in drawing auditory localization inward when synchronized with touch.

3 Discussion

The present findings suggest that the audiotactile magnet effect is not a general response to stimulation along the body’s central axis, but rather a phenomenon uniquely associated with the head. The illusion was strongest not only when touch was applied directly to the head, but also when tactile stimulation was delivered to a hand placed on the head. This indicates that it is not direct contact with the skin of the head that matters most, but rather the spatial and perceptual identity of the head itself as the location of stimulation.

These results support the idea that the head holds a special status in multisensory perception—not simply because of its anatomical position, but because it serves as a naturally activated stage for internalized sensory experience. The fact that stimulating a hand on the head produced a similar effect suggests that it is the head’s readiness to host imagined audiovisual events—rather than symbolic meaning or physical contact—that enhances the illusion.

How this effect relates to broader human tendencies toward in-head auditory localization—such as the propensity to internalize ambiguous or centered sounds—remains an open question for future investigation.

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